

Execution and transaction costs

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How a flat cost with full fills was replaced by a cost model that depends on spread, liquidity, volatility and order size; what the model is; what the studies around it found; and what it leaves out.

EVERY PERFORMANCE FIGURE IN THIS BRIEF IS A HISTORICAL BACKTEST. NO LIVE OR PAPER-TRADING RESULTS ARE SHOWN.

The execution-research problem

A backtest needs a rule for the price an order gets. The easiest rule, and the one in my first reported backtest of a systematic U.S. equity strategy that trades relative-value effects among single stocks, under portfolio-level risk controls, is that every order fills in full at the price that prompted it, for a small flat fee per round trip.

Real orders do three things the flat rule ignores. They cross part of the bid-ask spread. They move the price against themselves, more so the larger they are relative to the volume trading at the time. And they sometimes cannot be completed in the time available, so part of the expected profit is never earned. A flat cost hides all three, and it hides them unevenly: it charges the same at every account size, so it says nothing about capacity, and it flatters the least liquid names most.

The research question is therefore not what the cost is, but which conclusions survive realistic execution. How should cost be modeled when no fills at scale exist to measure it? Does a more sophisticated execution policy earn its complexity on data it was not tuned on? How does the answer change as the account grows? Paper trading cannot answer the first question: a paper account simulates its fills without sending orders to the market, so it cannot calibrate a cost model at any size.

The general modeling framework

Every component is textbook. What follows is the framework with each term explained; the settings used in the research are not disclosed.

Spread

Each side of a trade pays half of the bid-ask spread: a buyer pays above the mid price and a seller receives less. The spread is assumed from the stock's liquidity rather than observed per order. It is the one cost that does not scale with order size, so its share of total cost shrinks as the account grows.

Liquidity and volatility

Both enter through the impact term. Liquidity is the dollar volume available to trade against: the same order is a larger share of a thin name's volume than of a liquid name's. Volatility scales the price response: the same relative order moves a volatile stock further than a quiet one.

Market impact

Impact is modeled as a function of volatility and of order size relative to the volume available, following the square-root regularity reported across markets and order sizes: doubling an order less than doubles its impact. The coefficient that scales the term sets how fast cost grows with relative size and is the model's weakest input. Impact is charged as a temporary cost on each order; permanent impact is not modeled.

Partial fills and opportunity cost

When the volume available is thin, an order may fill only in part, and the model allows for that. The unfilled remainder is not assumed away: the profit it would have earned is carried as an opportunity cost, the component a flat cost cannot represent at all. This is also how cost comes to depend on account size: as orders grow relative to the volume available, impact rises first and, at much larger scale, unfilled shares become the largest loss.

Implementation shortfall

The accounting frame is Perold's implementation shortfall. Frictionless profit, every share at the decision price with no cost, less delay, spread, impact, fees and the opportunity cost of unfilled shares, equals executable profit, where delay is the price change between the decision and the order reaching the market. In the shortfall study the identity closed to the dollar on every trade. That checks the bookkeeping, not the level of any component, but it lets two order-handling rules be compared on the same trades, component by component.

Capacity

Capacity is not one number. It has at least three definitions that answer different questions: how much one signal in one name can absorb before fills or impact bind; how much the book can hold at once, which also depends on how often signals arrive and how long positions are held; and the economic scale at which the next dollar stops paying. In the capacity study the binding constraint differed across segments of the universe, available volume in some and modeled cost in others. Between half-years only the first segment to saturate repeated, and the levels moved by large factors, so I report no capacity figure.

What I tested and learned

More than a dozen studies ran on an intraday fill simulator over the trade record of the first reported backtest, which is superseded. A finding counted only if it held in at least three of four half-years and on both ordinary and unusually busy days, with any tuning done on the first half of the sample and the result read on the second.

Simplistic cost assumptions overstated confidence

The flat-cost result was not merely optimistic in level. It was uninformative about the two things that matter for implementation: how cost scales with size, and the shares that never fill. With the model in place, modeled cost rises with the simulated account and partial fills become material at larger scale. In the headline backtest at a \$1M simulated scale the modeled cost is about 40 basis points per round trip, and the ending multiple after that cost is 3.04×. Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85.

More complex policies did not improve held-out outcomes

- **Fill assumptions.** At small scale the fill assumption acts through unfilled shares, not through cost. Stressing it over a wide range changed captured profit very little except at the most restrictive setting, where more of each order went unfilled, and even that gap was unresolved at the base scale and resolved only narrowly at larger scale.
- **Working an order.** A schedule that works an order over a window kept more of the frictionless profit than a single immediate order, mostly through fewer unfilled shares. The lead held in every half-year but rests on a small share of trades, narrows under a less favorable fill-price convention, and is a statistical tie on unusually busy days. With standard errors that allow for dependence within sessions and within groups of related stocks the lead is only marginally significant; a coarser grouping, unreliable with so few groups, does not reach significance.

- **Optimal control.** The schedule was posed as a stochastic control problem with square-root temporary impact, a liquidity constraint and a penalty for unfilled shares; its value function satisfies a Hamilton–Jacobi–Bellman equation, solved here by backward induction and checked against closed forms, including the Almgren and Chriss linear-impact case. Whether to hurry or to spread turns on how fast the expected gain decays against how convex the impact cost is. The full solution spread most orders, because once it believed there was time and volume to finish, convex impact made spreading cheaper than hurrying. On data held out from tuning that lost to the best simple schedule, with a bootstrap interval that excludes zero, and adding even modest risk aversion drove the solution back toward the simple rule.
- **Learned timing.** A learned execution-timing policy, fitted Q-iteration and contextual bandits, did not beat a fixed schedule on held-out data. Across more than two hundred configurations its gain was matched by a copy trained on scrambled inputs and exceeded on random windows with no signal in them: it had rediscovered a fixed setting, not learned signal-specific behavior. One cell favored a learner on uncorrected intervals, and a correction for the number of configurations would remove it.

Realism changed research conclusions

Three conclusions moved. The headline figure fell, and I accepted the lower figure because the evidence behind it is more credible. The question of how large the strategy can run stopped having a single answer and became a question of which constraint binds where. And the appeal of a cleverer execution policy did not survive contact with held-out data: the simple rule, stress-tested, is what the simulation carries.

Live data as an additional calibration layer

Every cost figure in this brief is modeled; the impact coefficient comes from public tape data and a literature prior. Execution data from live operation are now being accumulated to refine and empirically calibrate those assumptions. A power study of what the measurement requires found that the benchmark price matters more than the number of fills: against the price at decision time, ordinary price movement over an order’s working window swamps the impact term, so the record must carry finer-grained benchmarks and deliberate variation in order size before the coefficient can be estimated with useful precision.

Limits

- **Where the studies ran.** The detailed studies used the first reported trade record, before the universe was rebuilt as of each date. Repeating them on the selected configuration’s trade record is a planned extension.
- **Busy days.** The busy-day half of the repeatability rule rests on a small number of sessions, so a pass or a fail there is weak evidence.
- **What the fill model leaves out.** Permanent impact, financing on exposure above equity, queue position, latency and crowding are not modeled. Where the fill model assumes an order completes, the finer-grained studies showed that at larger scale some did not, so modeled cost there is understated. The optimal-control study treated each interval’s volume as independent, and real volume persists.
- **Fill-price convention.** The fill price is a proxy for an executable price. In an earlier replay on the superseded universe an adverse bound on every fill removed essentially all of the simulated gain: a bound, not a forecast. That stress is part of the planned repeat on the selected configuration.
- **The headline backtest.** The 3.04× is one implementation of one two-year window. Its deflated Sharpe ratio is 0.85, a probability and not a reduced Sharpe, and a small fraction of its trades accounts for more than half of net profit. The methodology brief carries the full set of qualifications.

- **Authorship.** The research questions, methodology, system design, and interpretation are independently directed; implementation uses extensive AI-assisted development. The work has not yet had independent outside review.

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